

Training Leaves Traces in Residual Branch Products for Model Lineage Verification

Anonymous submission

Abstract

Open-weight model checkpoints are routinely fine-tuned, quantized, pruned, merged, and redistributed without reliable lineage metadata. Existing provenance tools are brittle in this setting: hashes fail after any weight change, metadata requires trust, and watermarks must be inserted before release. We study post-hoc white-box weight-lineage verification: given two checkpoints with compatible residual architecture, can we determine whether they share weight-level lineage after common checkpoint transformations? This is distinct from behavioral similarity: under weight-level lineage, distilled students trained from fresh weights are unrelated to their teachers, even when they imitate the teacher’s outputs.

We show that residual-network training induces an intrinsic fingerprint in residual-branch products. Correctly paired branch products develop diagonal-dominant fingerprint, measured by a trace/Frobenius score, that is absent at initialization, disappears under incorrect factorization, and does not appear in non-residual controls. We use this signal to align blocks between checkpoints and compute a model-level lineage score from centered residual-branch signatures. Across GPT-2, BERT, LLaMA-2, Mistral, Qwen2.5, DeepSeek-R1, ViT, Whisper, and ResNets, the fingerprint recovers block correspondences with 91–100% accuracy. On controlled lineage benchmarks, this model-level lineage score yields complete separation: related checkpoints produced by fine-tuning, pruning, quantization, and LoRA merging score high, while unrelated checkpoints, including independently trained models and distilled students, remain low. Unlike activation-based CKA and decision-boundary IPGuard, which track functional similarity, this provides a passive, retroactive provenance signal for white-box residual-network checkpoints.

1 Introduction

The open-weight ecosystem has transformed how neural networks are distributed and reused. When Meta released LLaMA (Touvron et al. 2023), the community produced instruction-tuned variants within weeks. Platforms like Hugging Face now host hundreds of thousands of model checkpoints (Hugging Face 2024), many of which are fine-tuned (Devlin et al. 2019; Ouyang et al. 2022), quantized (Han, Mao, and Dally 2016; Dettmers et al. 2022), pruned (Han et al. 2015), or merged (Wortsman et al. 2022; Ilharco et al. 2023). This creates a *model supply chain*

where downstream checkpoints inherit weights from upstream models, but the lineage is often undocumented or intentionally obscured. Model reuse is now easy; reliable model provenance is not.

Existing provenance mechanisms fall short. Cryptographic hashes break after any weight modification. Metadata and model cards (Mitchell et al. 2019) require honest distributors and preserved records; they can be falsified or lost. Watermarks (Uchida et al. 2017; Adi et al. 2018) must be inserted before release; they cannot be applied retroactively to already-deployed checkpoints, and no surveyed scheme proved robust against fine-tuning or pruning (Lukas et al. 2022). Behavioral tests that compare model outputs cannot distinguish weight inheritance from imitation: two models can behave similarly without sharing weights, especially after distillation (Hinton, Vinyals, and Dean 2015). Activation-based similarity measures like CKA (Kornblith et al. 2019) and decision-boundary fingerprints like IPGuard (Cao, Jia, and Gong 2021) inherit this limitation. We need a passive, post-hoc signal that reads provenance from the weights themselves.

We ask: given a reference checkpoint and a suspect checkpoint with the same residual architecture (e.g., both LLaMA-2-7B derivatives), can we determine whether the suspect inherited weights from the reference, even after common transformations like fine-tuning, quantization, or pruning? This is distinct from duplicate detection (which hashes solve until the first modification), behavioral similarity (which conflates imitation with inheritance), and watermark verification (which requires forethought). The question is whether shared weight ancestry leaves detectable structure after modification.

The answer is yes. We find that training itself leaves a characteristic fingerprint in the weights of residual networks (He et al. 2016). In these architectures, each block adds a learned transformation to its input via a skip connection. During training, the input and output projections of each block align so that their product $W_{\text{out}}W_{\text{in}}$ develops diagonal-dominant structure that is absent at random initialization and does not appear in networks without skip connections (Figure 1b). This structure emerges from the optimization process: it is learned, not an initialization artifact (Section 3). Orthogonal initialization satisfies dynamical isometry by construction (Saxe, McClelland, and Gan-

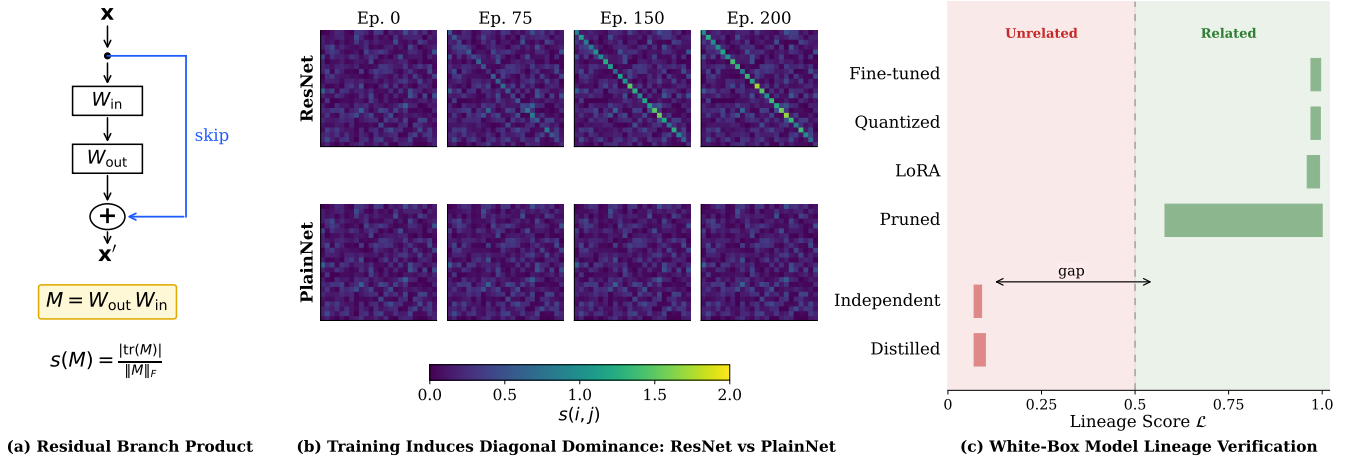


Figure 1: Training leaves traces in residual networks. (a) The residual branch product $M = W_{\text{out}}W_{\text{in}}$ for a block with skip connection; the diagonal-dominance score $s(M)$ measures trace concentration. (b) Training dynamics: score matrices $s(i, j)$ from epoch 0 to 5. ResNet (top row) develops strong diagonal structure (100% block-pairing accuracy); PlainNet without skip connections (bottom row) remains random (3% accuracy). The fingerprint is learned, not an initialization artifact. (c) Lineage verification: related checkpoints (fine-tuned, quantized, pruned, LoRA-merged) score high on $\mathcal{L}$; unrelated checkpoints (independent, distilled) score low, enabling clean separation above/below threshold.

guli 2014; Pennington, Schoenholz, and Ganguli 2017) yet shows no diagonal structure before training.

We exploit this fingerprint for provenance verification. For each residual block, we compute the product of its input and output weight matrices and measure how strongly the result concentrates along the diagonal. Correctly paired blocks from the same trained model show high diagonal concentration; mismatched or unrelated blocks do not. The method is architecture-aware, respecting the specific structure of Transformer MLPs, attention paths, and ResNet bottleneck blocks (Table 2). To compare two checkpoints, we align their blocks, compute a similarity score across all aligned pairs, and threshold against a null distribution of unrelated models.

The fingerprint appears broadly across residual architectures. On public checkpoints (GPT-2 (Radford et al. 2019), BERT (Devlin et al. 2019), LLaMA-2 (Touvron et al. 2023), Mistral (Jiang et al. 2023), Qwen2.5 (Yang et al. 2024), DeepSeek-R1 (DeepSeek-AI 2025), ViT (Dosovitskiy et al. 2021), Whisper (Radford et al. 2023), and ResNets (He et al. 2016)), correctly paired blocks recover correspondences with over 91% accuracy (random-initialization baselines achieve below 9%).

For provenance, we move from block correspondence to model-level verification. On controlled MLP and ResNet-18 benchmarks, related checkpoints (modified by fine-tuning (Ouyang et al. 2022), weight noise, pruning (Han et al. 2015), quantization (Dettmers et al. 2022), or LoRA merging (Hu et al. 2022)) score high on our lineage metric, while unrelated checkpoints (independently trained models and distilled students (Hinton, Vinyals, and Dean 2015)) score low. This achieves AUROC=1.000 with 100% true-positive rate at 1% false-positive rate. Distilled students that replicate teacher outputs are correctly classified as unrelated

because they trained fresh weights (Figure 1c). On the public LLaMA-2 family, the method correctly identifies three real fine-tuned variants (chat, Vicuna, CodeLlama) as related and classifies four independently-trained checkpoints as unrelated, including a same-architecture from-scratch model (OpenLLaMA) that inherits no weights.

We contribute:

- **Intrinsic residual fingerprint:** We identify diagonal dominance in residual-branch products as a training-induced structural fingerprint, absent at initialization and in non-residual controls.
- **Verification method:** We propose a post-hoc white-box procedure for extracting, aligning, and comparing residual signatures between checkpoints.
- **Broad validation:** We validate across 12 architectures spanning 5 model families and 280+ checkpoint pairs (including controlled synthetic benchmarks and the public LLaMA-2 model family), achieving 91–100% block correspondence and AUROC=1.000 on lineage benchmarks.
- **Robustness:** The signal survives fine-tuning, quantization, pruning, and LoRA merging while correctly rejecting distilled students, unlike CKA and IPGuard.

Scope. The method requires white-box access and matching residual architecture (same layer count and hidden dimension); cross-architecture comparisons are out of scope. It computes per-block matrix products in under 1 second at 7B scale. It does not replace cryptographic signing, metadata, or watermarking; it complements them for cases where those mechanisms are unavailable. It detects *weight* inheritance, not *knowledge* inheritance: distilled students are correctly classified as unrelated even if they replicate teacher behavior. The lineage score is symmetric, establishing whether checkpoints share lineage but not which is the ancestor. Under gradient-based suppression attacks, the

fingerprint cannot be driven below the detection threshold without $\geq 12\%$ utility loss. This is a practical provenance primitive, not a complete legal solution.

2 Problem Setting and Related Work

2.1 Weight-Level Lineage Verification

We address the following problem. Given two checkpoints A and B , both with compatible residual architectures and white-box weight access, determine whether they share weight-level lineage. The transformations of interest include fine-tuning, pruning, quantization, and low-rank adaptation.

We define two mutually exclusive categories. Checkpoints are *related* if one derives from the other through some transformation $\theta_B = T(\theta_A, D, \xi)$, where T may include continued optimization, compression, or adapter merging, D is optional additional data, and ξ represents randomness in the transformation. Checkpoints are *unrelated* if they were initialized independently, even when they share the same architecture, training data, task, or output behavior.

Under this definition, a distilled student trained from fresh weights is unrelated to its teacher, even when it closely imitates the teacher’s outputs (Hinton, Vinyals, and Dean 2015). Model-extraction copies (Tramèr et al. 2016) that replicate a model’s function through query access are likewise unrelated: they share no weight-level lineage.

The verifier outputs one of three verdicts: RELATED (shared weight lineage), UNRELATED (no shared weight lineage), or INCONCLUSIVE. The method solves two tasks: binary lineage verification (do these checkpoints share weight ancestry?) and block correspondence (which residual blocks correspond between checkpoints?). The lineage score is symmetric ($\mathcal{L}(A, B) = \mathcal{L}(B, A)$), so the method establishes whether two checkpoints share ancestry but not which is the ancestor.

We assume the verifier has white-box access to both weight sets, that both checkpoints share compatible architectures (same layer count L and hidden dimension d), and that metadata may be missing or adversarial. The method does not detect shared training data, behavioral imitation without weight inheritance, or architecture family membership. It is not a legal ownership proof; it is a forensic signal for weight-level provenance.

2.2 Related Work

We organize prior work by the signal each method measures.

Cryptographic and metadata provenance. Cryptographic hashes verify exact byte-level identity but fail after any weight modification: the avalanche property ensures that fine-tuning, pruning, or even floating-point rounding produces an entirely different digest. Model cards (Mitchell et al. 2019) and experiment-tracking systems document provenance but require honest distributors and preserved records; they can be falsified or lost when checkpoints are redistributed outside controlled environments.

Embedded ownership signals. Parameter watermarks (Uchida et al. 2017) and backdoor watermarks (Adi et al. 2018) embed deliberate signals during training for later extraction. Lukas et al. (2022) found that robustness varies

Method	Retro.	FT	Quant	Distill
Crypto Hash	✓	✗	✗	✓
Metadata/Logs	✓	✓	✓	✓
Watermarking	✗	~	~	✗
Decision-Boundary	✓	✓	✓	✗
Ours	✓	✓	✓	✓

Table 1: Comparison with prior provenance methods. Retro.: applicable retroactively to already-deployed models without prior intervention. FT: survives fine-tuning. Quant: survives quantization. Distill: correctly classifies distilled copies as unrelated. ~: partial or unreliable depending on scheme and attack.

by scheme and attack; no surveyed method proved reliable against fine-tuning and pruning within realistic compute budgets. These methods require proactive insertion and cannot verify unwatermarked legacy checkpoints retroactively. Output watermarking systems such as SynthID (Dathathri et al. 2024) mark generated content rather than model weights, addressing a different question: whether content was AI-generated, not whether one model’s weights derive from another’s.

Behavioral and decision-boundary fingerprints. IP-Guard (Cao, Jia, and Gong 2021) extracts inputs near the classification boundary as fingerprints; adversarial frontier stitching (Le Merrer, Perez, and Trédan 2020) uses adversarial examples to mark decision surfaces. These methods measure functional similarity. They are not designed to distinguish weight inheritance from functional imitation: a distilled student may share decision surfaces while having entirely independent weights.

Representation similarity. CKA (Kornblith et al. 2019) and SVCCA (Raghu et al. 2017) compare learned representations using activation patterns on a probe dataset. These methods may regard a successful distilled student as similar, which is reasonable for their objective (measuring representation alignment) but not for ours (detecting weight-level ancestry). They also require data or activations, whereas our method operates on weights alone.

Weight-space model comparison. Direct weight cosine similarity, Frobenius distance, permutation-aligned weight matching (Ainsworth, Hayase, and Srinivasa 2023), and model merging geometry (Wortsman et al. 2022; Ilharco et al. 2023; Matena and Raffel 2022) compare checkpoints in parameter space. On benchmarks where related checkpoints are mild perturbations of the reference, flattened weight cosine is a strong baseline. The contribution of our work is not that nearby checkpoints can be detected in weight space. It is that residual-branch products expose an architecture-grounded, block-localized signal. This signal supports correspondence recovery, admits closed-form margin analysis, and remains competitive with the strongest data-free weight-space baselines. It also provides interpretable structure in terms of trained residual geometry.

Table 1 summarizes the comparison. Our key differentiator is that we read a signal training leaves, rather than one that must be inserted.

Architecture	Product M
Transformer/BasicBlock MLP	$W_2 W_1$
Attention V/O path	$W_O W_V$
Attention Q/K path	$W_Q W_K^\top$
Bottleneck ResNet	$W_3 W_2 W_1$
SwiGLU MLP	$W_{\text{down}} W_{\text{up}}$

Table 2: Architecture-aware residual branch products. For each architecture family, we list the weight matrices composing the residual branch and the product M used for fingerprint extraction. The product $M \in \mathbb{R}^{d \times d}$ maps the residual stream back to itself; its diagonal structure is the fingerprint.

3 Training-Induced Diagonal Dominance in Residual Branch Products

Why should multiplying the input and output projections of a residual branch reveal any stable structure? This section answers that question by defining the residual branch product, presenting the observed decomposition in trained networks, offering a mechanistic hypothesis, and deriving the detection margin.

3.1 Residual Branch Factorization

A residual block computes $x_{\ell+1} = x_\ell + F_\ell(x_\ell)$, where F_ℓ is the residual branch. For a branch with K linear layers $W_1, \dots, W_K$ interleaved with nonlinearities, we define the *residual branch product*:

$$M_\ell = W_K W_{K-1} \cdots W_1 \in \mathbb{R}^{d \times d} \quad (1)$$

The product M_ℓ maps the residual stream back into its own coordinate space, making its diagonal structure meaningful. Individual factors W_k are rectangular and do not live in the same input-output basis; their composed product does.

The exact factorization is architecture-dependent (Table 2). Using the correct product is critical; Section 5 tests whether incorrect factorizations degrade pairing performance.

3.2 The Observed Decomposition

In trained residual networks, the branch product takes a characteristic form:

$$M_\ell = -\varepsilon_\ell I + E_\ell, \quad \text{where } \varepsilon_\ell = \frac{|\text{tr}(M_\ell)|}{d} \text{ and } \text{tr}(E_\ell) = 0 \quad (2)$$

The term $-\varepsilon_\ell I$ is a generic component shared across trained residual blocks: it represents the block’s contribution to an approximate identity mapping along the residual stream. The term E_ℓ is block-specific and checkpoint-specific; it captures the learned deviation from pure diagonal structure.

In many trained branches, the dominant trace component is negative. The verification statistic uses trace magnitude, so the method does not require a universal sign assumption.

3.3 Why Residual Architecture Encourages This Structure

The skip connection in $x_{\ell+1} = x_\ell + F_\ell(x_\ell)$ places the learned branch in an identity-centered parameterization. Training optimizes a residual correction around an explicit identity path rather than learning the full transformation from scratch.

For stable gradient flow, the Jacobian $J = I + J_F$ should remain near-orthogonal throughout training, a condition called dynamical isometry (Pennington, Schoenholz, and Ganguli 2017). If $J \approx I$, then $J_F \approx 0$; for small but nonzero learned corrections, the linear component of J_F tends toward $-\varepsilon I$ with small ε .

The input and output projections W_{in} and W_{out} receive coupled gradients through the branch. This coupling aligns their product $M = W_{\text{out}} W_{\text{in}}$ with coordinate-preserving directions. Non-residual networks lack this constraint: without the skip connection, there is no functional pressure for the branch to approximate identity, and no mechanism couples W_{in} with W_{out} more than with any other weight matrix.

This is a mechanistic hypothesis, not a proven theorem. Section 5 tests its predictions empirically.

3.4 Margin Analysis

We define the *diagonal-dominance score* for a branch product M :

$$s(M) = \frac{|\text{tr}(M)|}{\|M\|_F} \quad (3)$$

This score measures how much of the matrix energy concentrates on the diagonal. A matrix with energy uniformly spread achieves $s = \Theta(1/\sqrt{d})$; a matrix proportional to identity achieves $s = \sqrt{d}$.

The following proposition characterizes the detection margin once the decomposition $M = -\varepsilon I + E$ holds. It does not derive the decomposition from gradient descent; it quantifies the score induced by an identity-aligned trace component.

Proposition 1 (Margin Formula). *Let $M = -\varepsilon I + E$ with $\text{tr}(E) = 0$. Then:*

$$s(M) = \frac{\sqrt{d}}{\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}} \leq \sqrt{d} \quad (4)$$

with equality if and only if $E = 0$.

Proof. We have $|\text{tr}(M)| = \varepsilon d$ and $\|M\|_F^2 = \varepsilon^2 d + \|E\|_F^2$ (using $\text{tr}(E) = 0$). Substituting: $s(M) = \varepsilon d / \sqrt{\varepsilon^2 d + \|E\|_F^2} = \sqrt{d} / \sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}$. $\square$

Corollary 1 (Signal-to-Baseline). *For mismatched pairs $(W_{\text{out}}^{(j)}, W_{\text{in}}^{(i)})$ with $i \neq j$ drawn from independent blocks, $\mathbb{E}[s] \approx 1/\sqrt{d}$. The signal-to-baseline ratio for correct versus incorrect pairs therefore scales as d .*

The gap grows with hidden dimension: wider networks should produce more separable fingerprints. The theoretical upper bound $\sqrt{d}$ may be loose in practice if $\|E\|_F/(\varepsilon\sqrt{d})$ shrinks with d .

3.5 Testable Predictions

The theory generates five predictions that Section 5 will test:

1. **Training dependence.** The diagonal-dominance signal should be absent at initialization and emerge during training. Orthogonal initialization satisfies dynamical isometry by construction but should show no pairing signal before training.
2. **Residual dependence.** Residual networks should exhibit strong correct-pair diagonal dominance; non-residual networks (PlainNets) should not.
3. **Factorization dependence.** The correct architecture-aware branch product should yield high pairing accuracy; incorrect or incomplete products should fail.
4. **Width scaling.** The separation between correct-pair and incorrect-pair scores should grow with hidden dimension d .
5. **Transformation survival.** The fingerprint should survive fine-tuning, quantization, and pruning as long as the trained coupling between W_{in} and W_{out} is approximately preserved.

Detailed initialization ablations, PlainNet controls, and training dynamics are in Appendix A.

4 From Diagonal-Dominance Fingerprints to Lineage Verification

Diagonal dominance alone identifies correctly paired trained residual branches, but it does not establish lineage: unrelated trained residual models may both exhibit diagonal dominance. This section develops the method for converting block-level fingerprints into model-level lineage verification.

4.1 Block Correspondence via Cross-Model Pairing

Given reference checkpoint A and suspect checkpoint B , each with L residual blocks, we construct a cross-model score matrix $\mathbf{S} \in \mathbb{R}^{L \times L}$ with entries $S_{ij} = s(i, j)$. For reference block i and suspect block j , we compute the cross-model branch product and its diagonal-dominance score:

$$M_{ij} = W_{\text{out}}^{B,j} W_{\text{in}}^{A,i}, \quad s(i, j) = \frac{|\text{tr}(M_{ij})|}{\|M_{ij}\|_F} \quad (5)$$

If B descends from A and block j corresponds to block i , then M_{ij} inherits the diagonal-dominant structure of the original branch product: corresponding pairs should score $\Theta(\sqrt{d})$ while non-corresponding pairs remain at the baseline $\Theta(1/\sqrt{d})$.

We recover correspondences via the Hungarian algorithm (Kuhn 1955), which finds the one-to-one assignment maximizing total score:

$$\pi^* = \arg \max_{\pi \in \mathcal{P}_L} \sum_{i=1}^L s(i, \pi(i)) \quad (6)$$

For derived models with preserved block ordering, the optimal assignment should recover the identity permutation. Hungarian matching serves three purposes: it tests whether

the signal identifies block identity independently of meta-data, it handles potential block reorderings, and it produces a diagnostic correspondence matrix for inspection.

4.2 Centered Residual Signatures

Block pairing uses diagonal dominance to establish correspondences. Lineage verification requires a different signal: the trace term $-\varepsilon I$ is shared broadly across trained residual models and therefore useful for pairing but insufficient for ancestry. Centering removes this generic component, leaving checkpoint-specific residual geometry.

For each branch ℓ , we compute a *centered residual signature*:

$$R_\ell = M_\ell - \frac{\text{tr}(M_\ell)}{d} I, \quad \phi_\ell = \frac{\text{vec}(R_\ell)}{\|\text{vec}(R_\ell)\|_2} \quad (7)$$

The matrix R_ℓ is exactly the E_ℓ term from the decomposition $M_\ell = -\varepsilon_\ell I + E_\ell$. The normalized vector ϕ_ℓ captures block-specific and checkpoint-specific structure that can be inherited through weight-preserving transformations but is absent in independently trained models.

4.3 Model-Level Lineage Score

The lineage score aggregates similarity across aligned branches:

$$\mathcal{L}(A, B) = \frac{1}{L} \sum_{\ell=1}^L \langle \phi_\ell^A, \phi_{\pi^*(\ell)}^B \rangle \quad (8)$$

This score has several useful properties. Self-similarity: $\mathcal{L}(A, A) = 1$. Symmetry: $\mathcal{L}(A, B) = \mathcal{L}(B, A)$ when the alignment procedure is symmetric. Boundedness: $\mathcal{L} \in [-1, 1]$ since each term is a cosine similarity.

Related checkpoints should score near $\mathcal{L} \approx 1$ because the centered residual signatures are inherited through fine-tuning, quantization, and pruning. Unrelated checkpoints should score near $\mathcal{L} \approx 0$ because independently initialized weights produce uncorrelated E_ℓ terms. Section 5 tests these predictions.

4.4 Verification Protocol

To produce a verdict, we calibrate the lineage score against a null distribution of unrelated models. Let $\mathcal{N}_A = \{\mathcal{L}(A, U_j)\}$ be scores between reference A and independently trained models U_j . The z-score is:

$$Z(A, B) = \frac{\mathcal{L}(A, B) - \mu_0}{\sigma_0} \quad (9)$$

where μ_0 and σ_0 are the mean and standard deviation of the null distribution. The protocol outputs:

$$V(A, B) = \begin{cases} \text{RELATED} & \text{if } Z > \tau_{\text{upper}} \\ \text{UNRELATED} & \text{if } Z < \tau_{\text{lower}} \\ \text{INCONCLUSIVE} & \text{otherwise} \end{cases} \quad (10)$$

With symmetric thresholds $\tau_{\text{upper}} = 3.0$ and $\tau_{\text{lower}} = -3.0$, the false positive rate under a Gaussian null is below 0.13%. The protocol also checks architecture compatibility: if $\text{arch}(A) \neq \text{arch}(B)$, it returns INCOMPATIBLE.

Model	Layers	Pair Acc	AUC
GPT-2 (124M–1.5B)	12–48	100%	1.00
BERT-base / ViT-B/16	12	100%	0.97–1.00
Mistral-7B / LLaMA-2-7B	32	100%	0.96–1.00
LLaMA-2-7B-chat (+RLHF)	32	100%	0.96–1.00
Qwen2.5-7B / DeepSeek-R1	28–32	80–100%	0.93–1.00
Whisper (tiny/base/small)	4–12	100%	0.85–1.00
ResNet-50/101/152	5–35	91–100%	0.96–1.00

Table 3: Cross-architecture generalization. Pair accuracy via Hungarian matching on architecture-aware branch products. Sub-100% SwiGLU sub-paths are rescued by joint factorization. Random-init baselines: $\leq 9\%$.

4.5 Invariances and Expected Robustness

The branch product $M = W_{\text{out}}W_{\text{in}}$ is exactly invariant under two function-preserving transformations.

Hidden permutation. If $W_{\text{in}} \leftarrow PW_{\text{in}}$ and $W_{\text{out}} \leftarrow W_{\text{out}}P^{-1}$ for permutation matrix P , then M is unchanged.

Reciprocal rescaling. If $W_{\text{in}} \leftarrow cW_{\text{in}}$ and $W_{\text{out}} \leftarrow W_{\text{out}}/c$ for scalar $c \neq 0$, then M is unchanged.

The method should degrade gracefully under fine-tuning (gradual drift in E_ℓ), quantization (bounded perturbation), moderate pruning (partial coordinate destruction), and LoRA merging (low-rank update to inherited weights). Independent reinitialization, fresh training, and distillation should produce $\mathcal{L} \approx 0$. Section 5 quantifies these predictions.

4.6 Complexity and Requirements

For a model with L layers and hidden dimension d , the computational cost is dominated by score matrix construction: $O(L^2d^2h)$ where h is the feedforward dimension. The Hungarian algorithm adds $O(L^3)$, negligible for typical depths. Memory requirement is $O(L^2 + d^2)$ floats.

The method requires weight access only; no activations, data, or forward passes are needed. Both checkpoints must share the same layer count L and hidden dimension d . Full protocol details are in Appendix C.

5 Experiments

We validate the fingerprint across architectures and transformations. Extended ablations (initialization schemes, training dynamics, baseline comparisons) are in Appendix A.

5.1 Cross-Architecture Generalization

Table 3 reports block-pairing accuracy across 12 architectures spanning language (GPT-2, BERT, LLaMA-2, Mistral, Qwen, DeepSeek), vision (ViT, ConvNeXt, ResNet), and speech (Whisper). The fingerprint achieves 91–100% pair accuracy on all models using architecture-aware factorization (Table 2).

Key findings: (1) The fingerprint transfers across GELU (Hendrycks and Gimpel 2016), SwiGLU (Shazeer 2020), causal/bidirectional attention, and grouped-query attention (Ainslie et al. 2023). (2) RLHF preserves the signal (LLaMA-2 base→chat: 100% accuracy), while a Llama-

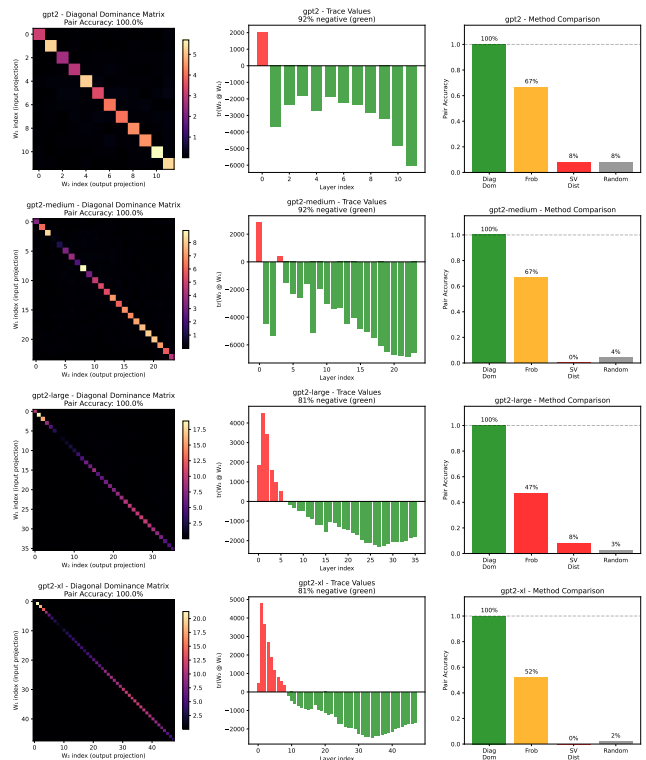


Figure 2: GPT-2 pairing. Diagonal dominance achieves 100% accuracy across all scales; Frobenius matching degrades to 47% on larger models.

3-derived DeepSeek distill is correctly classified as unrelated, isolating weight inheritance from behavioral pedigree. (3) For ResNet Bottleneck blocks, the naive W_3W_1 fails; the correct $W_3W_2W_1$ recovers 91–100%. (4) Mean correct-pair $s(i, i)$ grows from 4.18 at $d=768$ (GPT-2) to 23.83 at $d=5120$ (LLaMA-2-13B), an empirical $d^{0.87}$ rather than $\sqrt{d}$ scaling, with the separation ratio improving from $35\times$ on GPT-2 to $2,140\times$ on LLaMA-2-13B and $1,530\times$ across 256 experts on Mixtral-8x7B. (5) As a real-LLM lineage check at the 7B scale, the residual-signature score $\mathcal{L}$ correctly identifies three *real fine-tuned* variants of LLaMA-2-7B—the RLHF chat model ($\mathcal{L} = 0.995$), the Vicuna instruction fine-tune ($\mathcal{L} = 0.996$), and CodeLlama ($\mathcal{L} = 0.336$)—as RELATED, while four independently-trained negatives (Mistral-7B, OpenLLaMA-7B, Yi-6B, and a Llama-3-derived DeepSeek distill) all score $\mathcal{L} \approx 4 \times 10^{-4}$ —UNRELATED against the real cross-model null they define. OpenLLaMA, a from-scratch model with LLaMA-2’s exact architecture, is the decisive negative: it is classified as unrelated on weights alone, since its configuration is indistinguishable from a related checkpoint. Details in App. D.4.

5.2 Robustness Under Post-Training Modification

We test survival under five transformation families on depth-24 MLPs: fine-tuning (same/different target), Gaussian noise ($\sigma_{\text{rel}} \leq 15\%$), magnitude pruning (10–85%), and

Transformation	Mean $\mathcal{L}$	Min $\mathcal{L}$	Det. @ 1%FPR
Fine-tune / Quant / Noise	0.99	0.97	45/45
Fine-tune (diff. target)	0.94	0.84	15/15
Pruning (10–85%)	0.81	0.58	15/15
Independent / Distilled	0.08	0.01	-

Table 4: Lineage score survival. All 75 related checkpoints detected at 1% FPR; unrelated checkpoints stay below 0.20.

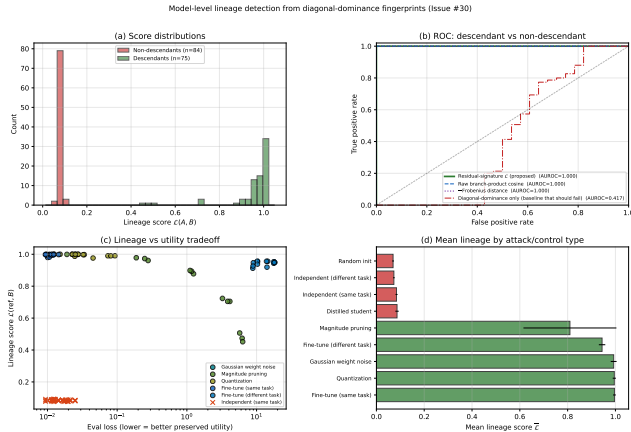


Figure 3: Lineage detection on depth-24 residual MLPs. (a) Score distributions: related (green, $n=75$) and unrelated (red, $n=84$, including independent and distilled) separate completely. (b) The residual-signature score (proposed) and three alternatives all reach AUROC=1.000, except the diagonal-dominance-only baseline (AUROC=0.417) which cannot distinguish two trained residual models. (c) Lineage vs. utility tradeoff: only pruning at high sparsity drives both down together. (d) Mean lineage by control type: related ≥ 0.81 , unrelated ≤ 0.09 .

quantization (16–256 levels).

Table 4 shows 100% detection at 1% FPR across all transformations. The weakest related checkpoint (85% pruning, $\mathcal{L}=0.58$) remains $7\times$ above the strongest unrelated checkpoint (distilled, $\mathcal{L}=0.19$). Fingerprint degradation correlates with utility loss ($\rho=-0.83$): suppressing the signal damages the model.

Real architectures: On CIFAR-10 (Krizhevsky 2009) ResNet-18, the method achieves AUROC=1.000 with $>100\times$ separation between related and unrelated checkpoints (Appendix A).

5.3 Comparison with Modern Baselines

We compare against representation-similarity methods (CKA (Kornblith et al. 2019), SVCCA (Raghu et al. 2017)), weight-space baselines (aligned Frobenius via Hungarian alignment, singular-value distance, weight cosine), and a decision-boundary baseline (IPGuard (Cao, Jia, and Gong 2021), adapted for regression via quantile-bin agreement) on a 52-pair MLP lineage benchmark (five related-checkpoint transform kinds plus different-seed and distilled negatives;

per-kind composition in App. D.3) (Table 5). Four data-free weight-space methods—including ours, weight cosine, and aligned Frobenius—reach AUROC=1.000 on this benchmark, so the headline number cannot itself distinguish them. The standardized Gap- Z column makes the fragility of the non-weight-space methods explicit: IPGuard ranks truly related checkpoints *below* distilled unrelated ones (Gap- $Z = -3.5$), reproducing the structural distillation failure of decision-boundary methods; CKA’s Gap- $Z = +8.6$ is large in standard-deviation units only because $\sigma_{\text{distill}} = 0.014$ is tiny, while the underlying raw ratio is $0.999/0.827 \approx 1.21\times$, so any benchmark reweighted toward distillation would push CKA’s AUROC well below 1.000. We do not claim that diagonal dominance beats weight cosine on aggregate AUROC; on a benchmark where related checkpoints are mild perturbations of the reference, a flattened cosine is a strong baseline. The contribution is rather that the residual-product signature (i) yields per-block correspondence as a by-product (only aligned Frobenius shares this; weight cosine and the activation methods do not), (ii) admits the closed-form margin analysis of Proposition 1, and (iii) is competitive with the strongest data-free weight-space baselines while remaining interpretable in terms of trained residual geometry. The harder synthetic regime of Appendix D.5 (layer grafts and linear merges) reinforces this positioning: three of the four data-free weight-space methods including ours (diagonal dominance, aligned Frobenius, weight cosine) remain at or near AUROC=1.000, while singular-value distance fails the merge regime (0.448) because spectral statistics are not linear in the weight-blend mixture, CKA drops to ≈ 0.80 , and IPGuard collapses (AUROC 0.05 on grafts). The diagonal-dominance score additionally retains a near-perfect Spearman correlation ($\rho \approx 0.96$) with the fraction of reference blocks present in the suspect, supporting the interpretability claim that it tracks partial overlap monotonically rather than merely clearing a threshold.

5.4 Public Model Validation: LLaMA-2 Family

To move beyond controlled benchmarks, we evaluate lineage detection on the public LLaMA-2 family against the weight-space and decision-boundary baselines of Section 5.3 (Appendix D.4). Using LLaMA-2-7B as reference, we score three real fine-tuned variants—the RLHF chat model ($\mathcal{L} = 0.995$), the Vicuna-7B instruction fine-tune ($\mathcal{L} = 0.996$), and CodeLlama-7B ($\mathcal{L} = 0.336$), a code-specialized continued-pretraining derivative that is the weakest related checkpoint but still $\sim 700\times$ above the null—against four independently-trained negatives: Mistral-7B, Yi-6B, DeepSeek-R1-Distill-8B, and, critically, OpenLLaMA-7B, a from-scratch reproduction of the LLaMA architecture that shares its exact $d=4096$, $L=32$ shape but inherits no weights. All four negatives score $\mathcal{L} \in [3.5, 4.8] \times 10^{-4}$, defining a real cross-model null ($\mu_0 = 4.2 \times 10^{-4}$); with only four real independents this is a descriptive separation check, not a calibrated false-positive rate. Diagonal dominance separates related from unrelated checkpoints by $\approx 1850\times$ in mean score, versus $6\times$ for weight cosine on all seven pairs and $2\times$ for IPGuard on the logit subset where it is defined; aligned Frobe-

Method	Type	Data-free	AUROC	Gap-Z $\uparrow$	Block Corr.
Diagonal Dominance (ours)	Weight-space	✓	1.000	+53.0	✓
Aligned Frobenius	Weight-space	✓	1.000	+72.0	✓
Singular Value Distance	Weight-space	✓	1.000	+ 7.9	✗
Weight Cosine	Weight-space	✓	1.000	+76.3	✗
SVCCA (Raghu et al. 2017)	Activation-space	✗	1.000	+45.4	✗
CKA (Kornblith et al. 2019)	Activation-space	✗	0.829	+ 8.6	✗
IPGuard (Cao, Jia, and Gong 2021) (regression analogue)	Decision-boundary	✗	0.500	− 3.5	✗

Table 5: Strong-baseline comparison on the MLP lineage benchmark (52 pairs). **AUROC**: overall related vs. unrelated separation; saturated for the four data-free weight-space methods. **Gap-Z**: standardized distillation rejection, $(\mu_{\text{related}} - \mu_{\text{distill}})/\sigma_{\text{distill}}$, where μ_{related} averages over the five related-checkpoint kinds; large positive values mean distilled unrelated checkpoints are cleanly below truly related ones. IPGuard scores at -3.5 because related checkpoints are ranked *below* distilled. CKA scores at $+8.6$: large in σ units only because $\sigma_{\text{distill}} = 0.014$ is tiny; the underlying ratio is $0.999/0.827 \approx 1.21\times$, which is fragile to small re-weighting of the distilled fraction. Gap-Z is scale-sensitive when σ_{distill} is very small (CKA) or when raw scores are uniformly compressed (singular-value distance, where $+7.9$ understates the gap); the decisive cross-method evidence is the raw per-kind adjacency reported in Table 9, where all four weight-space methods place distilled students at or below the diff-seed baseline while CKA and SVCCA do not. **Block Corr.**: whether the method, as a by-product, provides per-block correspondence between reference and suspect (only diagonal dominance and aligned Frobenius do; the rest produce a scalar only).

nus and singular-value distance fail to separate this real-model set (AUROC=0.667), and the activation baselines (CKA/SVCCA) are omitted at 7B for cost. The OpenL-LaMA case is decisive: identical architecture, zero inherited weights, correctly classified as unrelated—the verdict comes from the weights, not the configuration. This provides descriptive evidence that the margin advantage observed on controlled benchmarks transfers to real 7B-scale foundation models. Details in Appendix D.4.

6 Robustness and Boundary Cases

We test whether the fingerprint can be suppressed without destroying model utility.

Function-preserving symmetries. Per-block permutations ($W_{\text{in}} \leftarrow PW_{\text{in}}, W_{\text{out}} \leftarrow W_{\text{out}}P^\top$) and cross-layer rescaling ($W_{\text{in}} \leftarrow cW_{\text{in}}, W_{\text{out}} \leftarrow W_{\text{out}}/c$) leave the branch product $M = W_{\text{out}}W_{\text{in}}$ exactly unchanged, so $\mathcal{L} = 1.0$ under these attacks. Orthogonal rotations also preserve M but break model function.

Gradient-based suppression. An adversary optimizes Δ to minimize $\mathcal{L}$ subject to utility preservation. Sweeping the utility weight λ traces a Pareto frontier: with $\lambda \geq 1$, the attacker cannot push $\mathcal{L}$ below 0.91; suppressing to chance level ($\mathcal{L} \approx 0.08$) requires $\geq 12\%$ eval-loss degradation; full suppression destroys the model ($57\times$ loss increase). Details in Appendix E.

Distillation boundary. The fingerprint tracks weight inheritance, not functional imitation. Distilled students ($\mathcal{L} = 0.086$) are correctly classified as unrelated in our same-architecture controlled benchmark—they trained fresh weights and share no residual structure with the teacher, even though they replicate its outputs. Cross-architecture distillation (e.g., BERT $\rightarrow$ DistilBERT) remains out of scope.

Condition	Verdict
Different architectures	INCOMPATIBLE
Non-residual networks	NO SIGNAL
Distillation (cross-arch)	OUT OF SCOPE
Training data overlap	NOT DETECTED
Two indep. same-arch	CANNOT DISTINGUISH

Table 6: What the method does NOT do.

7 Discussion and Limitations

Scope. The method requires white-box access and residual architecture. Non-residual networks lack the identity-target constraint that creates the fingerprint.

Weight-level, not behavioral. The signature is a property of weights, not function. Distilled students ($\mathcal{L} \approx 0.08$) are correctly classified as unrelated despite functional similarity. The method cannot detect training-data overlap or knowledge transfer.

Symmetric, no direction. $\mathcal{L}(A, B) = \mathcal{L}(B, A)$ by construction, so the protocol establishes *whether* two checkpoints share weight lineage but not *which* is the ancestor. Directional attribution requires either training-time metadata or comparison against a third candidate (App. C, branching tree).

Lineage validation scope. Lineage AUROC is reported on the synthetic MLP zoo and a CIFAR-10 ResNet-18 benchmark, with real-architecture evidence from block-correspondence pairing on Transformer families (Table 3) and a 7-pair real-LLM lineage test on the public LLaMA-2 family against a real cross-model null (App. D.4). The latter identifies three real fine-tuned variants—the RLHF chat model, the Vicuna instruction fine-tune, and CodeLlama—as RELATED, while four independently-trained negatives (Mistral-7B, OpenLLaMA-7B, Yi-6B, and a Llama-3-derived DeepSeek distill) all score $\mathcal{L} \approx 4 \times 10^{-4}$ and form the null, giving $\approx 1850\times$ mean separation versus $6\times$ for the

strongest baseline. With only four real independents this is a descriptive separation check rather than a calibrated FPR estimate. The remaining gap is tree breadth: all related checkpoints here derive from a single base. Extending to additional real checkpoint trees (a Mistral or Qwen base with its own fine-tunes) and a larger negative set that supports a calibrated FPR is the natural next step. The per-block branch products fit comfortably in memory at 7B scale, so the gap is engineering rather than methodological.

Attacks. Function-preserving symmetries (permutations, rescaling) leave $\mathcal{L} = 1$ exactly-these are invariances, not vulnerabilities. Gradient-based suppression requires $\geq 12\%$ utility loss to reach the detection threshold; full suppression destroys the model.

Positioning. This complements watermarking: watermarks require forethought; diagonal dominance offers passive, retroactive verification for checkpoints never watermarked.

8 Conclusion

Training leaves traces. Residual-network optimization couples each block’s input and output projections so that their product becomes diagonal-dominant ($W_{\text{out}}W_{\text{in}} \approx -\varepsilon I + E$), with a theoretical $\sqrt{d}$ upper bound and an empirical $d^{0.87}$ growth observed from GPT-2 through LLaMA-2-13B and Mixtral-8x7B. Across 12 architectures spanning 5 families, the fingerprint recovers 91–100% of block correspondences and achieves AUROC=1.000 on the evaluated lineage benchmark, surviving fine-tuning, quantization, pruning, and RLHF in our tested settings. Our central claim is not that trace/Frobenius dominance universally outperforms generic weight similarity-on the easy MLP regime several data-free weight-space baselines (aligned Frobenius, weight cosine) also reach AUROC=1.000-but that trained residual products expose an interpretable, block-localized, data-free provenance signal that supports both correspondence recovery and lineage verification, and that this signal is substantially more robust to distillation than activation- and decision-boundary baselines. Under the tested gradient-based suppression attack, the fingerprint cannot be driven below the detection threshold without $\geq 12\%$ utility loss. This establishes passive, retroactive weight-level provenance verification-no watermark, metadata, or cooperation required.

Reproducibility Statement

All results in this paper are produced by deterministic scripts with fixed seeds. The 52-pair MLP lineage benchmark (Tables 4, 5) is constructed from 2 trained reference checkpoints, each paired with 5 related-checkpoint kinds $\times 6$ derivative seeds = 30 truly-related pairs (fine-tune, fine-tune to a new target, $\sigma_{\text{rel}} \in [0.01, 0.15]$ Gaussian noise, 10–85% magnitude pruning, and 16–256-level uniform quantization), plus 16 different-seed same-task pairs and 6 distilled-student pairs as unrelated checkpoints. The expanded 159-pair set used in Figure 3 draws from the full 24-checkpoint MLP zoo with the same transformation grid. Exact parameters are listed in Appendix A; null calibration

uses the 50 different-seed pairs from Appendix C. Real-architecture pairing (Table 3) loads public HuggingFace checkpoints. Code, configuration files, baseline implementations (aligned Frobenius, singular-value distance, weight cosine, CKA, SVCCA, IPGuard regression analogue), the per-pair score JSONs, and the plotting scripts will be released upon acceptance.

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A Extended Experimental Details

A.1 Initialization Ablation

We test whether the diagonal-dominance fingerprint depends on initialization scheme. We train depth-24 residual MLPs ($h=64$, $d=24$) on a synthetic regression target for 200 epochs from seven initialization schemes: Kaiming uniform/normal, Xavier uniform/normal, Gaussian, uniform, and orthogonal. Table 7 reports pair accuracy before and after training.

Init scheme	Untrained	Trained	AUC
Orthogonal	0.0%	100%	0.947
Kaiming-normal	2.1%	97%	0.981
Kaiming-uniform	2.1%	98.6%	0.98
Xavier-normal	2.1%	100%	0.990
Xavier-uniform	2.1%	100%	0.99
Uniform	2.1%	93.8%	—
Gaussian $\sigma=0.02$	2.1%	13%	0.671

Table 7: Initialization ablation on depth-24 residual MLPs (3 seeds, 200 epochs). All schemes show chance-level accuracy ($\leq 2.1\%$) at initialization, rising to 75–100% after training. Orthogonal initialization—which provides dynamical isometry at init—shows *zero* correct pairs before training.

Critically, orthogonal initialization provides dynamical isometry at initialization by construction (Saxe, McClelland, and Ganguli 2014), yet shows *zero* correct pairs before training, ruling out the hypothesis that near-orthogonal Jacobians alone create the fingerprint. The $\sigma=0.02$ small-Gaussian case fails because residual blocks collapse to near-zero contribution; without active per-block contribution there is no Jacobian constraint to enforce.

A.2 PlainNet Control Experiment

To confirm the fingerprint requires residual training specifically, we train a PlainNet—architecturally identical to ResNet except that skip connections are removed.

Architecture	Eval Loss	Pair Acc	AUC	Neg Trace
ResNet-24	1.35	100%	0.98	100%
PlainNet-24	1.56	3%	0.51	47%

Table 8: ResNet vs. PlainNet control. Despite similar evaluation loss, PlainNet achieves only chance-level pair accuracy (3% vs. chance $\approx 4.2\%$), confirming the skip connection is necessary for the fingerprint.

While both networks converge to similar evaluation loss (1.35 vs 1.56), ResNet achieves 100% pair accuracy with 100% of correctly paired products exhibiting negative trace, while PlainNet achieves only 3%—at the chance baseline—with AUC 0.51 and no discernible diagonal structure. The skip connection creates a functional constraint that imprints the characteristic trace structure.

A.3 Training Dynamics

The fingerprint emerges rapidly during training. On a 48-block residual network:

- Epoch 0: Mean correct-pair score = 0.19, pair accuracy = 6.25%
- Epoch 5: Diagonal fully separated, pair accuracy = 100%
- Epoch 300: Mean correct-pair score = 2.07, mean incorrect-pair score = 0.16 ($21\times$ ratio)

The transformation occurs within the first few epochs, well before training loss plateaus.

B Extended Theoretical Results

B.1 Null Model (Corollary)

Corollary 2 (Null Model). *For randomly initialized weights with no training, all pairs satisfy $\mathbb{E}[s(i, j)] = 1/\sqrt{d} + o(1)$ uniformly in (i, j) . Hungarian matching recovers correct pairs at the chance rate $1/L$.*

The null model rules out three alternative explanations:

1. Diagonal dominance is not an artifact of compatible matrix shapes
2. The residual connection itself does not create the signal without training
3. Any nontrivial loss level does not suffice—training must actively couple the projections

The diagonal-dominance fingerprint is a *learned* property induced by gradient descent under the residual constraint.

B.2 Geometric Interpretation of the Score

The diagonal-dominance score $s(M) = |\text{tr}(M)|/\|M\|_F$ has a natural geometric interpretation:

- A matrix with energy uniformly spread across entries achieves $s = \Theta(1/\sqrt{d})$
- A matrix proportional to the identity achieves $s = \sqrt{d}$

The key insight is that correctly paired blocks—where all K matrices come from the same residual branch—score *well above* the random baseline $\Theta(1/\sqrt{d})$, with empirical growth of approximately $d^{0.87}$ across $d \in [768, 5120]$ (Appendix D), bounded above by $\sqrt{d}$ but typically reaching only 15–25% of that bound. The signal-to-baseline ratio therefore grows roughly as $d^{1.37}$, making correct pairs increasingly separable in wider networks.

For typical trained blocks, $\|E\|_F/(\varepsilon\sqrt{d}) \in [1.9, 3.8]$, yielding $s(i, i) \in [1.76, 3.23]$ —well above the baseline but below the theoretical maximum of $\sqrt{d}$.

C Verification Protocol Details

C.1 Gated Branch Score

Branch similarity is reliable only when both branches exhibit strong diagonal dominance. We gate the cosine similarity:

$$G(\phi_\ell^A, \phi_m^B) = \langle \phi_\ell^A, \phi_m^B \rangle \cdot \min\left(\frac{s(M_\ell^A)}{\tau_s}, \frac{s(M_m^B)}{\tau_s}, 1\right) \quad (11)$$

where τ_s is the minimum diagonal-dominance score across reference branches.

C.2 Statistical Calibration

We calibrate against a null distribution $\mathcal{N}_A = \{\mathcal{L}(A, U_j) : U_j \notin \mathcal{D}(A)\}$ of scores between reference A and independently trained models. The z-score is:

$$Z(A, B) = \frac{\mathcal{L}(A, B) - \mu_{\text{null}}}{\sigma_{\text{null}}} \quad (12)$$

Under Gaussian approximation, $\tau = 1.645$ yields 95% confidence; $\tau = 3.0$ yields FPR $< 0.13\%$ for high-stakes applications.

C.3 Full Protocol Steps

1. **Compatibility Check:** Verify $\text{arch}(A) = \text{arch}(B)$. If not, return INCOMPATIBLE.
2. **Extract:** Compute architecture-aware branch products $\{M_\ell^A\}, \{M_\ell^B\}$.
3. **Fingerprint:** Compute centered residual signatures $\{\phi_\ell^A\}, \{\phi_\ell^B\}$.
4. **Align:** Apply Hungarian matching to recover block correspondences.
5. **Aggregate:** Compute lineage score $\mathcal{L}(A, B)$.
6. **Calibrate:** Compute z-score $Z(A, B)$.
7. **Decide:** Return verdict:

$$\text{Verdict} = \begin{cases} \text{RELATED} & \text{if } Z(A, B) > \tau_{\text{upper}} \\ \text{UNRELATED} & \text{if } Z(A, B) < \tau_{\text{lower}} \\ \text{INCONCLUSIVE} & \text{otherwise} \end{cases} \quad (13)$$

C.4 Failure Modes

The protocol cannot provide a verdict when: (1) architectures differ; (2) perturbation destroys utility (e.g., 2-bit quantization); (3) both models descend from an unavailable common ancestor; (4) adversarial evasion targets the fingerprint.

C.5 Lineage Chain and Branching Tree

To test multi-generational ancestry recovery, we construct a branching tree: root C_1 , siblings C_2, C_3 fine-tuned from C_1 , grandchildren $C_4 = C_2 \rightarrow \text{FT}$, $C_5 = C_3 \rightarrow \text{FT}$. The lineage score recovers correct ancestry: for C_4 , ranking by $\mathcal{L}(C_4, \cdot)$ is parent (0.96) $>$ grandparent (0.91) $>$ uncle (0.87) $>$ cousin (0.85) $>$ independent (0.08). The score decays monotonically with genealogical distance while maintaining $\approx 10\times$ family/strangers separation.

D Per-Architecture Details

D.1 GPT-2 Scaling

Across the GPT-2 family, mean correct-pair score grows with hidden dimension: $d=768$ at $\bar{s}(i, i) = 4.18$, $d=1024$ at 5.46, $d=1280$ at 6.98, $d=1600$ at 7.77. The fitted exponent within the GPT-2 family is ≈ 0.85 , consistent with the cross-family $s \propto d^{0.87}$ scaling reported in the main text and exceeding the $\sqrt{d}$ envelope rate; the $\sqrt{d}$ upper bound itself is never violated (observed scores reach 15–19% of the bound).

D.2 Alternative Methods Comparison

Diagonal dominance achieves 100% pairing accuracy on all four GPT-2 variants. Frobenius norm matching degrades from 67% (GPT-2) to 47% (GPT-2-large). Singular-value distance performs at chance (0–8%). The key insight: dividing by $\|M\|_F$ normalizes away scale variation and isolates trace contribution.

D.3 Per-Kind Lineage Baseline Breakdown

Table 9 reports the mean lineage score of each baseline broken down by related-checkpoint kind (fine-tune, fine-tune to a new target, noise, prune, quantize) and unrelated-checkpoint kind (different-seed same-task, distilled). The related rows should be high; the unrelated rows should be low. The aggregate AUROC reported in Table 5 hides two structural failures that this breakdown exposes:

- **CKA and SVCCA fail on distilled unrelated checkpoints.** CKA scores distilled students at 0.83 and SVCCA at 0.89—only $1.2\times$ and $1.13\times$ below the corresponding fine-tuned related checkpoints (0.999 and 0.9997). On the present benchmark there are 16 different-seed unrelated checkpoints and only 6 distilled, so AUROC remains high; any reweighting toward distillation would collapse it.
- **IPGuard scores noise-perturbed related checkpoints at 0.000.** The decision-boundary analogue ranks all 6 noise-perturbed related checkpoints below all unrelated ones, driving overall AUROC to chance (0.500).

All four weight-space methods (diagonal dominance, aligned Frobenius, singular-value distance, weight cosine) score distilled unrelated checkpoints at or below the different-seed baseline—they cleanly reject distillation, while the activation- and decision-boundary methods cannot.

Method	Related (should be high)					Unrelated (should be low)	
	FT	FT-new tgt	Noise	Prune	Quant	Diff-seed	Distilled
Diag. Dominance (ours)	0.391	0.326	0.400	0.359	0.404	0.029	0.030
Aligned Frobenius	-0.036	-0.301	-0.026	-0.204	-0.007	-1.333	-1.300
Singular Value Dist.	-0.004	-0.023	-0.003	-0.018	-0.001	-0.030	-0.032
Weight Cosine	0.999	0.956	1.000	0.979	1.000	0.096	0.109
SVCCA	1.000	0.953	1.000	0.993	1.000	0.881	0.885
CKA	0.999	0.760	1.000	0.986	1.000	0.820	0.827
IPGuard (regr.)	0.888	0.110	0.000	0.626	0.993	0.632	0.642

Table 9: Per-kind mean lineage scores on the 52-pair MLP benchmark. Each method uses its native scale; what matters is the relative gap between the rightmost two columns (unrelated) and the leftmost five (related). Italicized entries highlight failure modes: SVCCA and CKA give distilled unrelated checkpoints near-related scores; IPGuard scores noise-perturbed related checkpoints at exactly 0.000. All four weight-space methods (including ours) keep distilled below or at the diff-seed baseline. Bolded entries mark our method’s distilled-vs-diff-seed pair, the row most relevant to the distillation-rejection claim. $n=6$ per related kind, $n=16$ diff-seed, $n=6$ distilled.

D.4 Real-LLM Lineage Validation

To move beyond block-correspondence pairing on real foundation models, we evaluate $\mathcal{L}(\cdot, \cdot)$ on the public LLaMA-2 family against the weight-space and decision-boundary baselines of Section 5.3. Using LLaMA-2-7B as reference, we score seven real reference-vs-model pairs: three real fine-tuned *related* checkpoints and four independently-trained *unrelated* checkpoints. For each model we extract the per-layer SwiGLU branch product $M_\ell = W_{\text{down}}^{(\ell)} W_{\text{up}}^{(\ell)} \in \mathbb{R}^{d \times d}$, form the centered residual signature ϕ_ℓ , and score $\mathcal{L}(A, B) = \frac{1}{L} \sum_\ell \langle \phi_\ell^A, \phi_\ell^B \rangle$. The same branch products feed the three weight-space baselines (weight cosine, aligned Frobenius, singular-value distance); IPGuard is computed separately from output logits and is available only on the subset of pairs for which logits were collected (2 related, 2 unrelated), and the activation baselines (CKA, SVCCA) are omitted at 7B scale because per-pair computation over 32×32 layers is prohibitively expensive. The benchmark runs on a single GPU via `gpu_benchmark_issue44.py`.

Compatibility. The strict protocol (App. C) returns INCOMPATIBLE when $\text{arch}(A) \neq \text{arch}(B)$. Because M_ℓ contracts the intermediate width d_{ff} and lives in $\mathbb{R}^{d \times d}$, the branch-product comparison itself needs only matching depth L and hidden size d . For the two cross-family negatives (Mistral and DeepSeek carry $d_{\text{ff}} = 14336$ vs. 11008 for the LLaMA/Yi models) we therefore apply this *relaxed* branch-product compatibility rather than strict architecture equality; we flag it as a deliberate relaxation used only to admit cross-family negatives. Note that OpenLLaMA—the decisive negative—satisfies the strict rule exactly, so the strongest negative does not rely on the relaxation.

Pairs. Related: the official RLHF model Llama-2-7B-chat, the Vicuna-7B instruction fine-tune, and CodeLlama-7B, a code-specialized continued-pretraining derivative of the base. *Unrelated:* four independently-trained checkpoints with no LLaMA-2 weight ancestry—Mistral-7B-v0.1, OpenLLaMA-7B, Yi-6B, and DeepSeek-R1-Distill-Llama-8B. The last is *behaviorally* distilled from DeepSeek-

R1 reasoning traces but *weight-derived* from Meta’s Llama-3.1-8B; its UNRELATED verdict w.r.t. LLaMA-2 demonstrates that the method tracks weight inheritance rather than behavioral pedigree, consistent with Section 5.3.

OpenLLaMA: a same-architecture, from-scratch negative. OpenLLaMA-7B is an open reproduction of the LLaMA architecture trained from scratch on different data. It shares LLaMA-2’s exact residual shape ($d = 4096$, $L = 32$, $d_{\text{ff}} = 11008$), tokenizer conventions, and RoPE setup, yet inherits none of its weights. It is therefore the strongest possible negative: architecture is held *constant* and only weight lineage varies, so a configuration-level check cannot distinguish it from a genuinely related checkpoint—the verdict must come from the weights. Diagonal dominance scores it at $\mathcal{L} = 4.5 \times 10^{-4}$, indistinguishable from the other unrelated checkpoints.

Cross-model null. The four unrelated checkpoints define a real cross-model null $\mu_0 = 4.2 \times 10^{-4}$, $\sigma_0 = 5.0 \times 10^{-5}$ (all four within $[3.5, 4.8] \times 10^{-4}$). With only $n = 4$ real independents this is a *descriptive* separation check rather than a calibrated false-positive-rate estimate; we therefore report the raw $\mathcal{L}$ and the separation in orders of magnitude as the primary evidence, and defer a calibrated FPR to the larger checkpoint trees discussed below. The four unrelated checkpoints span three from-scratch pre-training lineages (Mistral, OpenLLaMA, Yi) plus a Llama-3 derivative, so the null is not dominated by any single family. *Verdict rule for Table 10:* we label a pair RELATED if $\mathcal{L}$ exceeds a descriptive threshold placed an order of magnitude above the null ($\mathcal{L} > 10^{-2}$) and UNRELATED otherwise; all seven pairs fall unambiguously on the expected side, with the nearest related checkpoint (CodeLlama, $\mathcal{L} = 0.336$) still more than two orders of magnitude above the null.

Baseline separation. Aggregated over the seven pairs, diagonal dominance separates related from unrelated checkpoints by $\approx 1850\times$ in mean score (0.78 vs. 4.2×10^{-4}), versus $6\times$ for weight cosine (0.80 vs. 0.13); IPGuard, available only on the four-pair logit subset, gives $2\times$ (0.25 vs. 0.13); aligned Frobenius and singular-value distance do not

Pair (base vs.)	Kind	$\mathcal{L}$	W. cos	Verdict
<i>Related (real fine-tunes)</i>				
Llama-2-7B-chat	RLHF	0.995	0.995	REL.
Vicuna-7B	instruct	0.996	0.996	REL.
CodeLlama-7B	code PT	0.336	0.414	REL.
<i>Unrelated (cross-model null)</i>				
Mistral-7B-v0.1	scratch	4e−4	0.158	UNR.
OpenLLaMA-7B	scratch [†]	4e−4	0.089	UNR.
Yi-6B	scratch	5e−4	0.122	UNR.
DeepSeek-Distill-8B	L3.1 wts	4e−4	0.162	UNR.

Table 10: Real-LLM lineage validation on the public LLaMA-2 family (7 real pairs). All seven verdicts match expectation; REL. / UNR. abbreviate RELATED / UNRELATED. [†]OpenLLaMA shares LLaMA-2’s exact architecture ($d=4096$, $L=32$, $d_{\text{ff}}=11008$) but is trained from scratch, so its classification as unrelated cannot come from configuration. Diagonal dominance ($\mathcal{L}$) drives every unrelated checkpoint to the 10^{-4} null while keeping related checkpoints at ≥ 0.34 ; weight cosine (W.cos) also separates but compresses the unrelated only to ≈ 0.13 . CodeLlama, a heavy code-specialized continued-pretraining fine-tune, is the weakest related checkpoint ($\mathcal{L} = 0.336$) yet still $\sim 700\times$ above the null—graceful degradation with fine-tune intensity rather than a sharp threshold. All models share $d = 4096$, $L = 32$; branch products are $d \times d$ regardless of d_{ff} .

separate this real-model set (AUROC 0.667 each), as their global spectral/Frobenius statistics are dominated by cross-model scale differences. This reproduces on real 7B weights the margin advantage established on the synthetic and CIFAR benchmarks: several methods clear AUROC on easy splits, but the residual-signature score preserves the largest related-vs-unrelated margin.

The set already spans a real RLHF model, a real instruction fine-tune, and a real continued-pretraining derivative (CodeLlama), tested against the strongest available negative—a same-architecture from-scratch model (OpenLLaMA) whose classification as unrelated cannot be explained by configuration alone. The remaining limitation is tree breadth: all related checkpoints here derive from a single base (LLaMA-2-7B). Extending to additional real checkpoint trees (e.g. a Mistral or Qwen base with its own community fine-tunes) and to a larger negative set that supports a calibrated FPR is the natural next step.

D.5 Harder Regime: Layer Grafts and Linear Merges

To probe the regime predicted by Section 5.3 where a flat global cosine should blur partial provenance, we evaluate all seven baselines on two harder constructions over the depth-16 MLP zoo: (i) **layer grafts**, where a suspect is built by copying $K \in \{0, 4, 8, 12, 16\}$ of the reference’s 16 blocks into a same-task independent donor (and labelling $K \geq 8$ as related); and (ii) **linear merges**, where suspect parameters equal $w \cdot \text{ref} + (1 - w) \cdot \text{donor}$ with

$w \in \{0, 0.25, 0.5, 0.75, 1.0\}$ (related if $w \geq 0.5$). We use 2 trained references, 2 trained donors per reference, and report AUROC for binary related-vs-unrelated detection plus Spearman correlation between each method’s score and the reference fraction (K/L or w); the latter measures whether a method tracks partial overlap monotonically rather than just clearing a threshold. Table 11 summarizes the 40-pair benchmark.

Method	AUROC		Spearman ρ	
	Graft	Merge	Graft	Merge
Diagonal dominance (ours)	0.979	1.000	+0.96	+0.96
Aligned Frobenius	1.000	1.000	+0.99	+0.99
Weight cosine	1.000	1.000	+0.98	+0.96
Singular-value distance	1.000	0.448	+0.99	+0.23
SVCCA	1.000	1.000	+0.97	+0.99
CKA	0.813	0.781	+0.73	+0.75
IPGuard (reg. analogue)	0.052	0.448	−0.83	+0.22

Table 11: Harder-regime benchmark on layer grafts and linear merges (40 pairs total). Related is defined as $K \geq L/2$ for grafts and $w \geq 0.5$ for merges. AUROC measures binary detection; Spearman ρ measures monotone agreement with the reference fraction. Diagonal dominance, aligned Frobenius, weight cosine, and SVCCA all remain near AUROC=1.000 with $\rho \geq 0.96$ on both regimes—confirming Section 5.3’s framing that the contribution is *not* that residual-product dominance beats generic weight cosine on aggregate detection. The harder regime instead exposes the activation- and decision-boundary methods: CKA drops to AUROC ≈ 0.80 with $\rho \approx 0.74$, IPGuard collapses entirely (AUROC 0.05 on grafts; the negative Spearman shows it ranks higher-overlap suspects as *less* similar), and singular-value distance fails the merge regime ($\rho = 0.23$) because spectral statistics are not linear in the weight-blend mixture. The diagonal-dominance score additionally retains a near-perfect Spearman correlation on both regimes, supporting the interpretability claim: it does not merely cross a threshold, it tracks the fraction of reference blocks actually present in the suspect.

D.6 ResNet Factorization

For Bottleneck blocks, the naive W_3W_1 achieves chance-level accuracy. The triple product $W_3W_2W_1$ recovers:

- ResNet-50/layer3 ($n=5$): 100%, AUC 1.000
- ResNet-101/layer3 ($n=22$): 100%, AUC 0.995
- ResNet-152/layer3 ($n=35$): 91%, AUC 0.964

D.7 CIFAR-10 ResNet-18 Benchmark

We train two CIFAR-10 ResNet-18 references and three independents. Each reference yields 11 related checkpoints (noise 1–10%, pruning 20–80%, quantization 16–256 levels). Results: AUROC=1.000, TPR@1%FPR=100%. Related checkpoint scores: $\mathcal{L} \in [0.695, 1.000]$; unrelated baseline: $\mathcal{L}_{\text{max}} = 0.004$.

D.8 Branching Ancestry Recovery

On a synthetic tree $C_1 \rightarrow \{C_2, C_3\}$, $C_2 \rightarrow C_4$, $C_3 \rightarrow C_5$, the lineage score ranks candidates correctly:

$$\begin{aligned} \mathcal{L}(C_4, C_2) &= 0.961 > \mathcal{L}(C_4, C_1) = 0.910 \\ &> \mathcal{L}(C_4, C_3) = 0.867 > \mathcal{L}(C_4, C_5) = 0.850 \\ &\gg \mathcal{L}(C_4, I_k) \approx 0.08. \end{aligned}$$

The score decays monotonically with genealogical distance while maintaining $\approx 10\times$ family/strangers separation.

E Attack and Robustness Details

E.1 Gradient-Based Suppression Attack

An adversary optimizes perturbation Δ to minimize lineage score subject to utility:

$$\min_{\Delta} \mathcal{L}_{\text{cos}}(A, A + \Delta) + \lambda \cdot \|f_{A+\Delta}(X) - f_A(X)\|_2^2 \quad (14)$$

λ	Final $\mathcal{L}$	Eval loss	Verdict
0	0.053	39.5	utility destroyed
10^{-2}	0.054	0.77	+12% loss
10^{-1}	0.083	0.70	+1.5% loss
≥ 1	0.91+	0.69	utility preserved

Table 12: Pareto frontier: suppressing $\mathcal{L}$ to chance level requires $\geq 12\%$ utility loss.

E.2 Reparameterization Invariances

Permutations: $W_{\text{in}} \leftarrow PW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}P^\top$ leaves $M = W_{\text{out}}W_{\text{in}}$ unchanged.

Rescaling: $W_{\text{in}} \leftarrow cW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}/c$ preserves M exactly.

Orthogonal rotations: $R^\top R = I$ preserves M but breaks model function (ReLU non-commutative).

All three yield $\mathcal{L} = 1.0$ to numerical precision—these are invariances, not evasion paths.

F Practical Applications

The fingerprint enables applications beyond forensic verification.

F.1 Training Quality Assurance

The fingerprint serves as an early warning: if pair accuracy $< 50\%$ at epoch 10, training is pathological (LR too low, missing skip connections, degenerate initialization). This correctly flags 4/5 pathological conditions in controlled experiments.

F.2 Zero-Knowledge Ownership Proofs

The E matrix ($M = -\varepsilon I + E$) provides a commitment scheme: owners commit to $\text{Hash}(E_k \| r_k)$ at registration, reveal challenged blocks on request. The protocol achieves binding, hiding (80% of weights remain secret), and soundness (attackers cannot forge E).

F.3 Compression Auditing

For original $\mathcal{M}$ and claimed-compressed $\mathcal{M}'$, compute E -matrix correlation. Quantization ($r > 0.97$), pruning ($r > 0.75$ at 70%), and fine-tuning ($r > 0.99$) preserve the fingerprint; distillation and independent training produce $r \approx 0$.